

MEDASANI PUNEETH

AI/ML Systems Engineer - Edge AI & Robotics - Full-Stack AI Applications

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PROFILE

AI/ML engineer and robotics systems builder with hands-on experience spanning embodied AI and physical systems, edge AI deployment (ESP32/Jetson), autonomous robotic systems (ROS/SLAM), and full-stack AI applications (RAG pipelines, LangGraph, ChromaDB).

TECHNICAL SKILLS

Languages: Python, C/C++, Java, SQL

AI / ML: PyTorch, Scikit-learn, Deep Learning, ML, CNNs, LSTMs, Transformers, RAG pipelines, LangGraph, ChromaDB, FAISS

Robotics & Systems: ROS, ROS-Gazebo, SLAM, UAV Development

Edge & Hardware: FPGA (Xilinx), NVIDIA Jetson Nano, Raspberry Pi, Arduino, ESP-32

Dev & Tools: FastAPI, Next.js, Docker, Git, DAMASK simulation suite

Prototyping: PCB Design, 3D Modelling & Printing

WORK EXPERIENCE

Research Engineering Intern | University of Hyderabad (Materials Science Lab)

Automated a high-fidelity computational pipeline using DAMASK for steel-rolling simulations (multi-pass, re-gridding), reducing manual setup and execution overhead.

Engineered a streamlined data extraction and normalization architecture to transform raw simulation outputs into structured, ML-ready datasets.

ENGINEERING PROJECTS

Multi-AI Stack Research Assistant (RAG System)

Architected a production-grade AI research authoring assistant using LangGraph, FastAPI, Next.js, ChromaDB, and NVIDIA NIM APIs, enables semantic paper retrieval via vector embeddings and AI-assisted thesis editing.

Designed multi-agent orchestration layer with retrieval-augmented generation (RAG) pipeline indexing 1,000+ research papers, reduced literature review time.

AI Professor Assistant — Computer Vision System

Built a dual-function CV system: automated attendance tracking using face recognition at 97% accuracy

AI-powered answer sheet evaluation pipeline reducing manual grading effort.

Autonomous Drone Platform — Aerial Surveillance & CV

Designed and built a custom UAV platform for unmanned surveillance, integrated a server-side computer vision pipeline processing live video at 25 FPS.

Implemented flight control firmware and sensor fusion for autonomous waypoint navigation and vision pipeline.

ROS-Gazebo SLAM Mobile Robot

Simulated, built and deployed a differential-drive mobile robot in ROS-Gazebo with SLAM integration for real-time 2D mapping and localization.

Implemented sensor fusion across LiDAR, camera and IMU feeds using an extended Kalman filter for robust pose estimation.

Edge AI Pothole Detection & Infrastructure Monitoring System

Designed an edge AI anomaly detection system using accelerometer data to identify potholes by intensity and frequency signatures, reports and geo-maps incidents for municipal infrastructure monitoring.

Deployed on ESP-32 with on-device inference.

Vision based cooking robot

Automated cooking robot, computer vision ingredient monitoring, Arduino motor control, YAML-driven recipes, and a live web control dashboard.

Architected a modular system for an autonomous cooking platform incorporating Computer Vision for ingredient monitoring.

Engineered a YAML-driven recipe engine that translates high-level cooking instructions into low-level motor commands.

AI-Integrated Bio-Wearable | Edge AI Health Monitoring System

Engineered an ESP32-based wearable integrated with bio-sensors to monitor SpO2, heart rate, and temperature in real-time for isolated patient care.

Developed and deployed a lightweight anomaly detection model directly on the edge to identify physiological distress and automate emergency alerts to healthcare providers.

COMPETITIONS & RECOGNITION

Google Big Code Challenge 2026 — Qualified Round 2 out of Indian competitive programming pool

IITB Maze Solving Robot Competition — Won regional round, got qualified for national.

Mahindra Line Following Competition — Finished Second Place

Xilinx Adaptive Computing Challenge — Won Hardware Prize

AI Professor Assistant — Inter-College Challenge — CV attendance + grading system; competed against 20+ teams.

DIY Arduboy Challenge — Designed hybrid game emulation device, awarded free hardware by organizers

AI Wearable Vital Signs Monitor — End-to-end wearable health device with real-time edge inference; presented at university showcase

Completed Harvard CS50: Introduction to Computer Science — edX certified

EDUCATION

Integrated Master of Technology | University of Hyderabad | Expected July 2029 (Currently pursuing second year)

Relevant focus: AI/ML Systems, Robotics, Edge Computing, Computer Vision

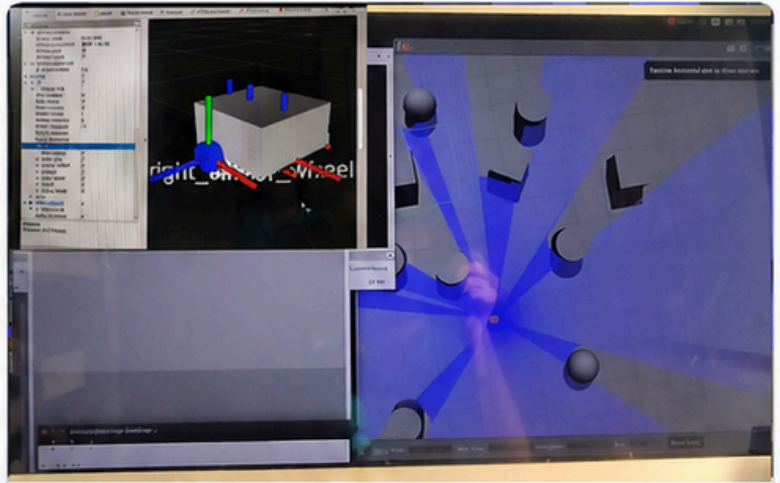
<Robotics & Edge AI> <Systems Portfolio>

EMBEDDED SYSTEMS • COMPUTER VISION • AUTONOMY • REAL-WORLD APPLICATIONS



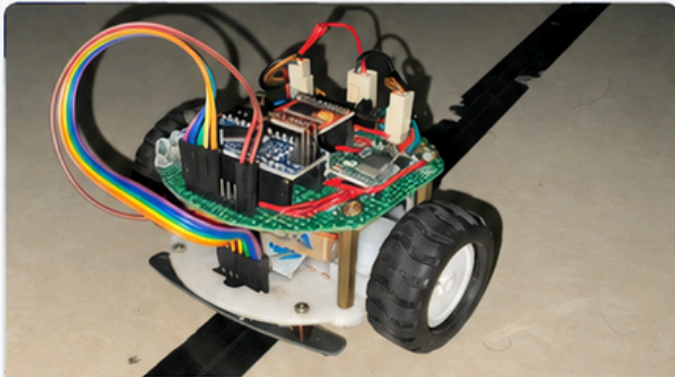
AUTONOMOUS DRONE PLATFORM

- Aerial surveillance & computer vision
- Custom UAV with autonomous navigation, sensor fusion & real-time video processing



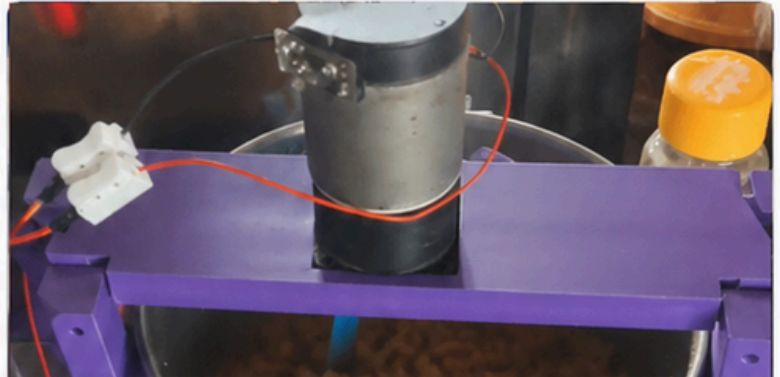
ROS-GAZEBO SLAM

- 2D mapping & localization
- Simulated & deployed differential-drive robot with LiDAR & IMU fusion using EKF



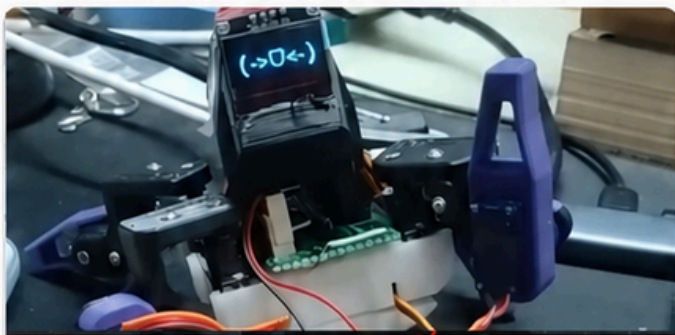
MAZE SOLVER ROBOT

- Autonomous navigation
- Solves complex mazes using sensor data, exploration algorithms and shortest path planning



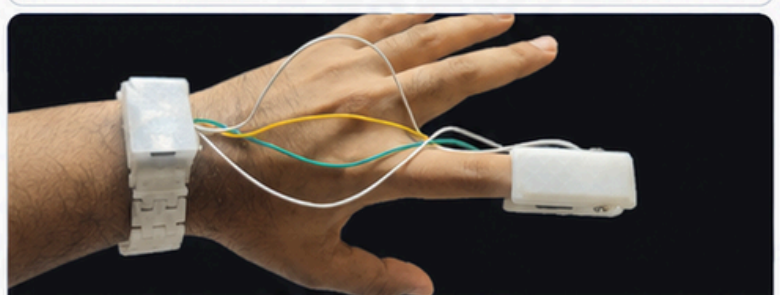
VISION-BASED COOKING ROBOT

- Computer vision + YAML recipes + automation
- Monitors ingredients using CV, executes YAML-driven recipes and provides live web control dashboard for a fully autonomous cooking system



ROBOTIC ARACHNID

- 4-legged mobile robot
- Bio-inspired walking mechanism with servo actuation for stable locomotion over rough terrain



WEARABLE HEALTH MONITOR

- ESP32-based wearable with SpO2, heart rate, and temperature monitoring
- Edge anomaly detection for real-time alerts in isolated patient care